JAN–R–184

31 JULY 1945

JOINT ARMY-NAVY SPECIFICATION

RESISTORS, FIXED, WIRE-WOUND (LOW-POWER)

Army Number
71–4904

Navy Number
17R20

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy.

A. APPLICABLE SPECIFICATIONS.

A–1. The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–P–14—Plastic-Materials, Molded Thermosetting.
JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN–P–107—Packaging and Packing for Overseas Shipment—Boxes, Wood, Wire-Bound.
JAN–P–116—Packaging and Packing for Overseas Shipment—Preservation, Methods of.
JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

U. S. ARMY SPECIFICATION

100–2—Standard Specification for Marking Shipments by Contractors.

SIGNAL CORPS TENTATIVE SPECIFICATION

72–6–15—Marking of Interior Containers (for Signal Corps Equipment).

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[1]

B. CLASSIFICATION.

B–1. This specification covers low-power, fixed wire-wound, axial-lead resistors which are for use primarily in communications and electronic equipment, and which have power ratings of ½, 1, and 2 watts, with a resistance tolerance of ±5 and ±10 percent, as specified in the contract or order.

[1] Applicable only to Navy purchases.

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B–2. *Styles.*—Resistors covered by this specification shall be of the styles shown on figures 4 to 6, inclusive, as specified in the contract or order. (See par. H–2.)

B–3. *Characteristics.*—Resistors shall be of the following characteristics as specified in the contract or order:

Characteristic:

A—Resistant to salt water immersion.

B—Resistant to moisture.

C—Resistant to both salt water immersion and moisture.

C. MATERIAL AND WORKMANSHIP.

C–1. *Material.*—The material for each part shall be as specified herein. Where a definite material is not specified, a suitable material shall be used. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

C–1a. *Plastic material.*—All plastic material when used in the construction of these resistors shall be molded plastic, conforming to the requirements of Joint Army-Navy Specification JAN–P–14, type MTS–E–3, or better, except that the water absorption, and the power factor at 1 megacycle may be 0.1 and 0.02, respectively.

C–2. *Substitution of material.*—If the resistor manufacturer desires to substitute for a specified or previously approved material or fabricated part, he shall submit to the bureau or agency concerned, with his bid or sample, a statement to that effect, describing the proposed substitution, together with evidence to substantiate his claims that such substitute is suitable. Similar notification and substantiating evidence shall be submitted at any later time that substitution becomes necessary or desirable. At the discretion of the bureau or agency concerned, test samples may be required to prove the suitability of the proposed substitute.

C–3. *Soldering fluxes.*—Only noncorrosive fluxes shall be used unless it can be shown that corrosive elements have been satisfactorily removed. After soldering, all excess flux shall be removed. Soldering shall not be used primarily for obtaining mechanical strength. Electrical connections shall be mechanically secure and electrically continuous before and after soldering.

C–4. *Iron and steel.*—The use of ferrous materials (see par. D–1e) for current-carrying parts is prohibited.

C–5. *Workmanship.*—All resistors shall be manufactured and processed in a careful and workmanlike manner, in accordance with good design and practice.

D. GENERAL REQUIREMENTS.

D–1. *Definitions.*—

D–1a. *Still air.*—Still air is air with no circulation other than that created by the heat of the resistor being operated.

D–1b. *Free-space mounting.*—Free-space mounting shall mean horizontal mounting of the resistors with no object closer than 3 inches to the protective coating, except the mounting base which shall not be closer than 2 inches below the resistors. To attain this condition, the resistors shall be soldered to light-weight terminals.

D–1c. *Hot spot.*—The hot spot is the point or elemental area of maximum temperature on the external surface of a resistor.

D–1d. *Thermal stability.*—Thermal stability has been reached when

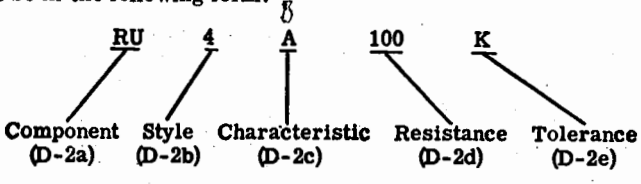
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there is no appreciable change between two successive measurements taken at 5-minute intervals.

D–1e. *Ferrous materials.*—Ferrous materials, for the purpose of this specification, are those materials containing more than 40 percent iron.

D–1f. *Nonstandard resistors.*—Nonstandard resistors are resistors whose styles are not covered by figures 4 to 6, inclusive, and whose maximum resistance values exceed those shown on figures 4 to 6, inclusive. (See par. H–3.)

D–2. *Type designation.*—The type designation of fixed wire-wound resistors (low power) furnished in accordance with this specification shall be in the following form:



D–2a. *Component.*—Fixed wire-wound resistors (low power) are identified by the symbol "RU."

D–2b. *Style.*—The resistor style is designated by a 1-digit number as shown on the applicable figure identifying a resistor of a particular physical shape, size, and power rating.

D–2c. *Characteristic.*—The characteristic is identified by a 1-letter symbol indicating the characteristic of the resistor in accordance with paragraph B–3.

D–2d. *Resistance.*—The resistance expressed in ohms is identified by 3 digits, the first 2 of which represent significant figures and the last digit the number of zeros which follow the significant figures. The letter "R" may be substituted for 1 of the digits to represent the decimal point, but when R is used the remaining digits of the group represent significant figures as in the following example:

*R*56=.56 ohms.

D–2e. *Tolerance.*—The symmetrical resistance tolerance, in percent of total resistance value, is denoted by a letter as follows:

J indicates ±5.
K indicates ±10.

D–2e (1). Unless specifically required by the contract or order, resistors having the K tolerance shall not be furnished for military use.

E. DETAIL REQUIREMENTS.

E–1. *Design and construction.*—Resistors shall consist of forms wound with a single layer of resistance wire and covered with protective coatings. Resistors shall be wound with a continuous unbroken length of conductor without joints, welds, bonds, etc., excepting at end terminals. The wire shall possess a uniform cross-sectional area and be as free as practicable from particles of impurities, grain growth, or other factors contributing to spot weakness. The design shall be such as to preclude shorting of turns and to obtain a minimum voltage between adjacent turns.

E–1a. *Windings.*—The cross-sectional area of the resistance wire shall be the maximum possible consistent with the other requirements

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herein, and in no case shall the minimum diameter be less than 0.0015 inch nominal (0.0014 inch absolute minimum).

E–1b. *Terminals.*—Resistor terminals shall be securely and permanently connected, both electrically and mechanically, to the resistance element preferably by soldering, welding, or brazing. They shall be suitably coated so that the leads may be easily soldered with rosin-core solder consisting of 40 percent tin and 60 percent lead. Wire-lead lengths and diameters shall be as shown on the applicable figures. The terminals shall suffer no breakage when twisted or pulled as outlined in paragraph F–16.

E–2. *Resistance.*—Resistors shall have a direct-current resistance within the specified tolerance of the nominal resistance. (See par. F–5.)

E–3. *Dielectric strength.*—There shall be no flashover, nor any evidence of break-down of insulation or change in resistance in excess of 1 percent as a result of the application of the test potential specified in paragraph F–6.

E–4. *Temperature cycling.*—Resistors shall be capable of withstanding without mechanical injury the temperature cycling outlined in paragraph F–7. The permanent change in resistance of the resistors shall be not more than 3 percent.

E–5. *Short-time overload.*—Resistors shall be capable of withstanding without damage the application of the test potential outlined in paragraph F–8 and shall not change in resistance by more than 2.5 percent.

E–6. *Effect of soldering.*—As a result of the test specified in paragraph F–9, resistors shall show no mechanical damage and shall not change in resistance by more than 1 percent.

E–7. *Vibration.*—Resistors shall suffer no mechanical damage and shall not change in resistance by more than 1 percent as the result of the vibration test outlined in paragraph F–10.

E–8. *Salt-water-immersion cycling (applicable to characteristics A and C).*—As a result of the test outlined in paragraph F–11 resistors shall show no damage to any part of the resistors and shall withstand an average in excess of 9 cycles of immersion without changing in resistance by more than 10 percent, between the initial resistance measurement and any succeeding measurements made at the end of each cycle. Only 1 of each group of 10 resistors tested may withstand less than 5 cycles and this resistor shall be excluded when the group average is computed.

E–9. *Abbreviated salt-water immersion.*—As a result of the test outlined in paragraph F–12, the resistors shall show no damage and shall not change in resistance by more than 2 percent.

E–10. *Moisture resistance (applicable to characteristics B and C).*—When subjected to the test outlined in paragraph F–13, resistors shall show no evidence of mechanical damage; nor shall there be a resistance change in excess of 3 percent; nor shall the insulation resistance be less than 10 megohms at the initial measurement or at the measurement made at the end of 10 cycles.

E–11. *Resistance change with temperature.*—The percent change in resistance at any temperature between the range of −55°C. (−67° F.) and 110° C. (230° F.) referred to an ambient temperature of 25° C. (77° F.) shall not be more than ±0.025 percent per degree centigrade

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(±0.014 percent per degree Fahrenheit) for resistors over 10 ohms and ±0.050 percent per degree centigrade (±0.028 percent per degree Fahrenheit) for resistors 10 ohms and less. (See par. F–14.)

E–12. *Power rating.*—Resistors operated in still air and free space at an ambient temperature of 40° C. (104° F.) shall dissipate not less than the rated power input for the styles shown on the applicable figures without exceeding a temperature rise of 70° C. (126° F.). Resistors operated continuously at temperatures other than 40° C. (104° F.) shall be capable of dissipating not less than that portion shown on Figure 1, of their rated nominal wattage without exceeding the maximum allowable hot spot temperature of 110° C. (230° F.). (See par. F–15.)

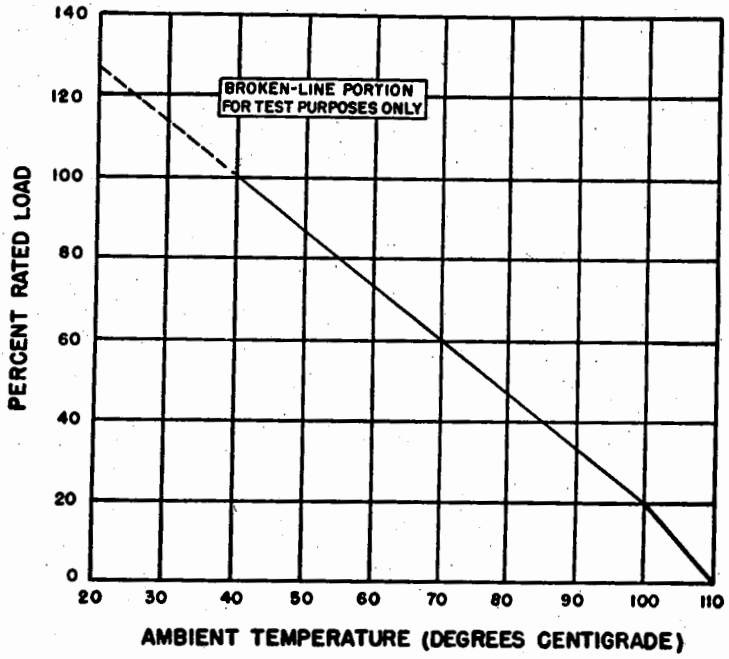


FIGURE 1.—Power rating curve for continuous duty.

E–13. *High temperature load-life.*—The resistors shall not change in resistance by more than 5 percent as the result of the test outlined in paragraph F–17.

E–14. *Overall resistance change.*—As a result of tests as outlined in paragraphs F–6 to F–10 in test group II of table II, or F–6 to F–9 in test group III of table II, the total change in resistance between the initial measurement as outlined in paragraph F–6 and the final measure-

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ment as outlined in paragraphs F–9 and F–10 shall be not more than 5 percent.

E–15. *Marking.*—Resistors shall be color-coded by bands of color in accordance with the requirements outlined in figure 2 and table I to indicate their type, resistance value and resistance tolerance.

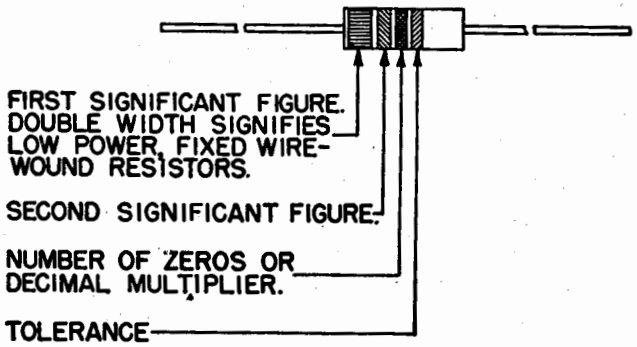


FIGURE 2.—Color coding.

TABLE I

Color	Numeral or number of zeros	Decimal multiplier	Tolerance
			Percent
Black	0		
Brown	1		
Red	2		
Orange	3		
Yellow	4		
Green	5		
Blue	6		
Violet	7		
Gray	8		
White	9		
Gold		0.1	±5 (J)
Silver		.01	±10 (K)

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Classification of tests.*—The methods of sampling, inspection, and tests conducted upon resistors shall be classified as follows:

Qualification (type or brand approval) tests (at a laboratory designated by the bureau or agency concerned).
Inspection tests (at place of manufacture).
Acceptance tests (generally at place of manufacture).[1]

F–2. *Standard test conditions and methods.*—

F–2a. *Temperature and humidity.*—Unless otherwise specified, all measurements and tests shall be made at 25° C. (77° F.) and at ambient humidity. If it is necessary to make such measurements at

[1] Applicable only to Army purchases.

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other temperatures, the measurements shall be corrected at the discretion of the Government inspector.

F–2b. *Methods and precautions.*—In the measurement of resistance, care shall be taken that not more than 10 percent of rated load be applied to the resistors at any time.

F–3. *Qualification (type or brand approval) tests (at a laboratory designated by the bureau or agency concerned).*—For qualification (type or brand) approval, tests shall be made on a sample quantity to insure that the basic design produces resistors meeting all the requirements of this specification.

F–3a. *Selection and number of samples.*—A set of 30 sample resistors of the highest resistance value for each type of wire used, wound with the smallest size wire (see par. E–1a), shall be submitted in the particular style for which qualification (type or brand) approval is desired. In addition, a set of 20 10-ohm resistor samples shall be submitted in the particular style for which qualification (type or brand) approval is desired.

F–3b. *Test routine.*—Sample resistors shall be subjected to the tests in the order listed in table II. The 30 sample resistors shall be subjected to the tests of test group I and then divided equally into three groups, II to IV, inclusive, and subjected to the tests for their particular groups. The 20 sample resistors of 10 ohms each shall be subjected to the tests of test group I and then divided equally into two groups, II and III, and subjected to the tests for their particular groups.

TABLE II.—*Outline of tests.*[1]

Test	Paragraph	Test	Paragraph
TEST GROUP I (ALL SAMPLES) TO QUALIFY ALL CHARACTERISTICS		TEST GROUP III (10 SAMPLES FROM EACH SET) TO QUALIFY CHARACTERISTICS B AND C	
Visual and mechanical inspection	F–4.		
Resistance	F–5.	Dielectric strength	F–6.
		Temperature cycling	F–7.
TEST GROUP II (10 SAMPLES FROM EACH SET) TO QUALIFY CHARACTERISTICS A AND C		Short-time overload	F–8.
		Effect of soldering	F–9.
		Moisture resistance	F–13.
Dielectric strength	F–6.		
Temperature cycling	F–7.	TEST GROUP IV (10 SAMPLES FROM EACH SET) TO QUALIFY ALL CHARACTERISTICS	
Short-time overload	F–8.		
Effect of soldering	F–9.		
Vibration	F–10.	Resistance change with temperature	F–14.
Salt-water-immersion cycling	F–11.	Power rating	F–15.
		Terminal strength	F–16.
		High-temperature load-life	F–17.

[1] See paragraph E–14.

F–3c. *Failures.*—One failure shall be permitted in each of tests groups II, III, and IV of Table II, with a total of not more than 3 failures permitted for each set of 50 samples. For the purpose of qualification, the results of the salt-water-immersion test outlined in paragraph F–11 shall not be applied to this paragraph.

F–4. *Visual and mechanical inspection.*—Resistors shall be inspected to verify that the physical dimensions, construction, and marking are

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as specified, and that the workmanship is satisfactory. (See pars. C–5, E–1, and E–15.)

F–5. *Resistance.*—The resistance of each resistor shall be measured and compared with the specified nominal resistance. (See par. E–2.)

F–6. *Dielectric strength.*—Resistors shall be placed in the trough of a 90-degree metallic V-block of such size that the resistor, when centered, shall not extend beyond the end of the V-block.

F–6a. *Atmospheric pressure.*—An approximate sine-wave test potential of 700 volts r. m. s. for ½-watt resistors and 1,000 volts r. m. s. for 1- and 2-watt resistors from an alternating-current supply at commercial-line frequency shall be applied for 1 minute between the terminals connected together and the V-block. (See par. E–3.)

F–6b. *Reduced pressure.*—With the resistor held in an atmosphere having a pressure equivalent to 3.4 inches of mercury an approximate sine-wave test potential of 500 volts r. m. s. for ½-watt resistors and 700 volts r. m. s. for 1- and 2-watt resistors from an alternating-current supply at commercial-line frequency shall be applied for 1 minute between the terminals connected together and the V-block. (See par. E–3.)

F–7. *Temperature cycling.*—Resistors shall be subjected to the temperature cycle shown in table III for a total of 5 cycles performed continuously, one cycle following the other, and shall be held in the specified temperatures for the length of time shown in table III. The rate of temperature change within the climate chamber shall be not less than 2° C. (3.6° F.) per minute. The resistors may be transferred from one chamber to another in which case they shall be kept at room temperature for the times shown in table III for 25° C. (77° F.) between exposures to the extreme temperatures. The resistance shall be measured before cycling and 1 hour after the fifth cycle. (See par. E–4.)

TABLE III.—*Temperature cycle*

	Degrees Centigrade	Degrees Fahrenheit	Time
Start at	25°±5°	77°±9°	
Raise to	85°+3°−0°	185°+5.5°−0°	30 minutes.
Return to	25°±5°	77°±9°	10 to 15 minutes.
Reduce to	−55°+0°−3°	−67°+0°−5.5°	30 minutes.
Return to	25°±5°	77°±9°	10 to 15 minutes.

F–8. *Short-time overload.*—Following a measurement of resistance, resistors shall be subjected to a potential corresponding to three times rated wattage for 5 seconds, after which the resistance shall again be measured. (See par. E–5.)

F–9. *Effect of soldering.*—Resistor terminals shall be immersed one at a time, for 3 seconds each, in a pot of molten solder at a temperature of approximately 350° C. (662° F.) to a point ⅛ inch from the resistor body. The resistance shall be measured after the resistors have remained for 30 minutes at room temperature. (See par. E–6.)

F–10. *Vibration.*—

F–10a. *Mounting.*—For this test, the terminals of the resistors shall be soldered to rigid mounts or terminal lugs. The spacing of the mounts shall be such that the length of each resistor lead shall be approximately ⅜ inch when measured from the edge of the supporting terminal to the resistor body.

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F–10b. *Procedure.*—Resistors shall be mounted in such a position as is most likely to cause failure. Following a measurement of resistance, resistors shall be subjected to a simple harmonic motion having an amplitude of 0.03 inch (maximum excursion 0.06 inch), the frequency being varied uniformly between the approximate limits of 10 and 55 cycles per second. The entire range of frequencies from 10 to 55 cycles and return shall be traversed in approximately 1 minute. This motion shall be applied for a total period of 5 hours. Resistance shall then be measured. (See par. E–7.)

F–11. *Salt-water-immersion cycling (applicable to characteristics A and C).*—

F–11a. *Conditioning.*—Prior to immersion in salt water, the resistors shall have rated load applied at an ambient temperature of 40° C. (104° F.) for a period of 1½ hours. Voltage shall then be removed and after 30 minutes the resistance shall be measured at this same ambient temperature.

F–11b. *Procedure.*—After conditioning as outlined in paragraph F–11a, the resistors shall immediately be immersed for 1 hour in a bath of saturated salt water maintained at 100°±4° C. (212°±7.2° F.). Within 5 seconds after removal from this bath, resistors shall be immersed for a period of 1 hour in another bath of saturated salt water maintained at 0°±4° C. (32°±7.2° F.). Resistors shall then be removed, thoroughly and quickly rinsed in tap water, wiped dry with a soft cloth, and rated voltage applied for 1 hour at an ambient temperature of 40° C. (104° F.). Voltage shall then be removed and resistance measured after 30 minutes at 40° C. (104° F.). The salt-water bath specified herein shall be made up using common table salt (NaCl). The baths shall be of sufficient volume to maintain relatively constant temperature for the duration of the tests.

F–11c. *Cycling.*—Resistors shall be subjected to the cycling procedure of paragraph F–11b for a total of not more than 14 cycles performed continuously, one cycle following the other. The cycling of any individual resistor shall be halted if at the end of any cycle the change in resistance from the initial resistance measurement of paragraph F–11a has exceeded the change permitted by the applicable requirement. (See par. E–8.)

F–12. *Abbreviated salt-water-immersion test.*—Immediately following the temperature-cycling test outlined in paragraph F–7, the resistors shall be measured for resistance and then immediately subjected to the following immersion test. Immerse the resistors for 1 hour in a bath of saturated salt water maintained at a temperature of 100°±4° C. (212°±7.2° F.). Within 5 seconds after removal from this bath, resistors shall be immersed for a period of 1 hour in another bath of saturated salt water maintained at 0°±4° C. (32°±7.2° F.). Upon removal from this second bath, the resistors shall immediately be given another cycle of immersion (both hot and cold baths). Upon removal from the second bath (cold), the resistors shall be immediately thoroughly and quickly rinsed in tap water, wiped dry with a soft cloth, and rated voltage applied for ½ hour at room temperature. The voltage shall then be removed and the resistance measured after 1 hour at room temperature. The salt-water baths specified herein shall be made up using common table salt (NaCl). The baths shall be of sufficient volume to maintain relatively constant temperature for the duration of the tests. (See pars. E–9 and F–19b (2).)

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F–13. *Moisture resistance (applicable to characteristics B and C).*—The moisture resistance test shall be performed in accordance with Figure 3 and the following:

F–13a. *Mounting.*—The terminals of the resistors shall be soldered to rigid mounts or terminal lugs. The spacing of the mounts shall be such that the length of each resistor lead shall be approximately ⅜ inch when measured from the edge of the supporting terminal to the resistor body.

F–13b. *Measurement (prior to first cycle).*—At the end of the initial drying period, and as shown on figure 3, the resistance of each resistor shall be measured.

F–13c. *Test procedure.*—Resistors shall be subjected to the cycle as shown on figure 3 for a total of 10 continuous cycles.

F–13d. *Steps 1 and 4.*—During the first two hours of each of these steps rated load, derated to 55° C. in accordance with figure 1, shall be applied as shown on figure 3.

F–13e. *Daily step 7.*—During step 7, at least 1 hour but not more than 4 hours after step 7 begins, the resistors shall be removed from the humidity chamber in order to perform steps 7a and 7b. After performance of steps 7a and 7b, the resistors shall be returned to the humidity chamber and kept there until the next cycle begins.

F–13e (1). *Step 7b.*—Within 15 minutes after the completion of step 7a, the resistors shall be subjected to the vibration test as outlined in paragraph F–10 except that the duration of this test shall be as shown on Figure 3. (See par. E–10.)

F–14. *Reistance change with temperature.*—The resistance of each resistor shall be measured at a temperature of 25° C. (77° F.). The resistors shall then be maintained for 30 minutes at each of the ambient temperatures in the order listed in table IV. At the end of each 30-minute period the resistance of each resistor shall be determined at the specified temperature maintained during the period. The temperature adjustment shall be accurate to within 1° C. (1.8° F.). The percent change in resistance shall be computed in accordance with the following equation:

$$\frac{\Delta R}{R25} \times 100 = \text{percent change in resistance}$$

Where

ΔR = change in resistance between 25° C. (77° F.) and test temperature.

$R25$ = resistance at 25° C. (77° F.)

(See par. E–11.)

TABLE IV.—*Resistance-change temperatures*

Degrees centigrade	Degrees Fahrenheit	Degrees centigrade	Degrees Fahrenheit
0	32	50	122
−55	−67	75	167
−15	5	100	212
25	77	110	230

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F–15. *Power rating.*—Resistors shall be supported in still air and free space. Rated wattage shall be applied to each resistor in accordance with paragraph E–12, until thermal stability has been reached. The hot-spot temperature shall then be determined by the use of a thermocouple.

F–15a. *Thermocouple.*—Resistors shall be horizontally mounted during this test with the thermocouple junction resting on the hot spot and the thermocouple wires hanging down on opposite sides of the resistors. Weights of at least 2 ounces on each lead shall provide the pressure of the thermojunction against the surface of the resistor under test. The thermocouple wire shall have a diameter of 0.0004 inch (No. 38AWG) or smaller.

F–16. *Terminal strength.*—Resistors shall be firmly clamped, and a direct pull shall be gradually applied to each terminal, one at a time, in the direction away from the resistor and along the longitudinal axis until a load of 5 pounds is reached. Each terminal shall then be bent through 90 degrees at a point ¼ inch from the body of the resistor, and rotated through 360 degrees in alternating directions for 5 rotations at the rate of approximately 5 seconds per rotation. (See par. E–1b.)

F–17. *High-temperature load-life.*—

F–17a. *Test conditions.*—This test shall be conducted at an ambient temperature of 85°±5° C. (185°±9° F.) for a total of 500 hours. Resistors shall be soldered to light-weight terminals during this test; the effective length of each lead shall be 1 inch. Resistors shall be so arranged that the temperature of any one resistor shall not appreciably influence the temperature of any other resistor. There shall be no undue draft over the resistors. The voltage used shall be the rated working voltage, and adequate precaution shall be taken to maintain constant voltage on resistors under test.

F–17b. *Procedure.*—Rated voltage in accordance with figure 1 from a direct-current supply shall be applied intermittently, 1½ hours on and ½ hour off, for a total of 500 hours. (See par. E–13.)

F–18. *Test equipment and inspection facilities.*—The resistor manufacturer shall furnish and maintain all necessary facilities and equipment for making all inspection and acceptance tests. The test equipment shall be adequate in quantity to enable inspection tests to keep up with production. The test equipment shall be of sufficient accuracy and quality to permit performance of the required tests. Where practicable, the inspection of resistors shall be made at the place of manufacture.

F–19. *Inspection tests.*—Inspection tests consist of mutually exclusive groups of test, termed production and production-sampling tests, which are performed by the manufacturer under the supervision of the Government inspector, to fully establish the adequacy of the material presented to the Government inspector. The grouping of the tests is dependent upon importance, complexity, destructiveness, and test equipment availability.

F–19a. *Production tests.*—Production tests are tests made on all resistors.

F–19a (1). *Test routine.*—All resistors shall be subjected to the production tests in the order listed in table V.

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TABLE V.—*Production tests*

Test	Paragraph
Resistance	F-5.
Visual and mechanical inspection	F-4.

F-19a (2). *Resubmission of failures.*—No resistor failing the production tests shall be submitted for retest unless it is accompanied by a statement of what action has been taken by the manufacturer to correct the defect causing the original rejection.

F-19b. *Production-sampling tests.*—Production-sampling tests, as outlined in table VII shall be divided into two groups: A, as described in paragraph F-19b (1), and B, as described in paragraph F-19b (2).

F-19b (1). *Production-sampling, group A tests.*—Selection of samples and criteria for lot rejection. Samples shall be selected at random, in accordance with table VI, by the Government inspector from lots of resistors that have passed the production tests. If more than the specified number of failures shown in the table occur, the lot from which the samples were chosen shall be withheld until a remedial treatment or retest satisfactory to the Government inspector is performed.

F-19b (1) a. *Lot.*—A lot is to be construed as meaning a group consisting of resistors of the same style, characteristic, wire size, and wire type, even though differing in resistance values, produced under uniform conditions. The lot shall preferably consist of resistors from the same natural unit of production, such as shift's production, day's production, etc. Groups of resistors likely to differ widely in quality because of differences in raw material, manufacturing methods, or test methods, should not be included in the same lot.

TABLE VI.—*Number of samples and failures permitted (group A only)*

Lot size	Sample quantities			Maximum allowable number of failures—			
				In first sample quantity		In first and second sample quantities combined	
	First sample quantity	Second sample quantity	First and second samples combined	For any one test	For all tests combined	For any one test	For all tests combined
150–499	50	100	150	2	3	4	9
500–799	50	100	150	2	3	4	9
800–1,299	75	150	225	3	5	5	11
1,300–3,199	100	200	300	3	6	8	16
3,200–7,999	150	300	450	5	9	13	23
8,000–21,999	200	400	600	6	11	16	32
22,000 and over	300	600	900	8	16	25	47

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F–19b (1) b. *Test routine.*—The production-sampling tests specified under group A shall be performed in the order listed in table VII.

F–19b (2). *Production-sampling, group B tests, selection of samples.*—If more than 1,000 resistors of any one style and grade are produced during the course of a week, at least once every week, the manufacturer shall permit the Government inspector to select 6 resistors per 1,000, but not over 25 in each style and grade, that have passed all production tests.

F–19b (2) a. *Failures.*—If any sample fails, the manufacturer shall take steps to find the cause of the failure and prove to the Government inspector that appropriate remedial measures have been taken before submitting later production of that type of resistor for inspection. Shipments shall not be delayed pending results of this test.

F–19b (2) b. *Test routine.*—The tests specified under group B in table VII, shall be performed on samples chosen in accordance with procedure outlined in paragraph F–19b (2).

TABLE VII.—*Production-sampling tests*

Test	Paragraph
GROUP A ONLY	
Dielectric strength	F–6.
Short-time overload	F–8.
Visual and mechanical inspection	F–4
GROUP B ONLY	
Temperature cycling	F–7.
Abbreviated salt-water-immersion	F–12.

F–19b (2) c. *Disposition of samples.*—The sample resistors which have been subjected to the temperature-cycling and abbreviated salt-water-immersion tests as outlined in paragraphs F–7 and F–12, shall be destroyed or stamped "Unfit for Military Use." Destruction shall be witnessed by the Government inspector. The stamping shall be of a permanent nature.

F–20. *Acceptance tests (applicable to Army purchases only).*—Acceptance tests are tests made by or under the observation of the Government inspector to determine whether the inspection tests have been properly conducted by the manufacturer and to assure that material accepted is of a sufficiently high level of quality. At the discretion of the Government inspector, acceptance tests may consist of observation of the inspection tests performed by the manufacturer or, in addition, these tests may be repeated by or under the observation of the Government inspector. Where the Government inspector elects to observe the performance of the factory tests, the selection of samples shall be made by him or under his supervision. The extent of the acceptance tests shall be so varied as to reduce the inspection load when quality of material presented is high. Statistical Quality-Control acceptance methods shall be employed insofar as is possible by the

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Government inspector and material found unsatisfactory by these methods shall be returned to the manufacturer for repair or retest before resubmission to the Government inspector. Prior to use, the Quality-Control method should be coordinated with the manufacturer by the inspecting agency.

F–21. *Additional tests and inspection requirements.*—Tests, not necessarily detailed in the applying specifications, but required to determine definite conformity therewith, shall be conducted as may be required to satisfy the Government inspector concerned that the resistors being produced or offered for acceptance are in full accord with the requirements of the specification and are entirely suitable for the purposes intended.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G–1. *Packaging.*—

G–1a. *Packaging for domestic or overseas shipment.*—Unless otherwise specified, resistors shall be shipped in bulk (initial installation in equipment) or as units (stock spares).

G–1a (1). *Bulk shipments.*—No packaging beyond that normally provided under best commercial practice will be required.

G–1a (2). *Unit shipment.*—Resistors shall be individually packaged in accordance with the requirements of Method IA of Joint Army-Navy Specification JAN–P–116. No preservation compound shall be applied. A quantity of packages shall then be further packaged in an intermediate container. The intermediate container shall be a reverse tuck, straight tuck, or two-piece telescope style folding carton. The gross weight shall not exceed approximately 5 pounds.

G–2. *Packing.*—

G–2a. *For domestic shipment.*—Unless otherwise specified, the subject commodity, packaged in bulk or in unit and intermediate packages as specified in paragraphs G–1a (1) and G–1a (2), respectively, shall be packed in suitable style corrugated or solid fiberboard boxes having a minimum Mullen test of 200 pounds. All flaps shall be sealed by means of taping, stitching, or sealing with a suitable adhesive or by a combination of these methods. If taped, not less than 2-inch 60-pound basis weight kraft gummed tape shall be used. The gross weight shall not exceed approximately 50 pounds.

G–2b. *For overseas shipment.*—Unless otherwise specified, the subject commodity, packed for domestic shipment (see par. G–2a) shall be overpacked in cleated plywood boxes, nailed wood boxes, or wirebound boxes conforming to the requirements of Joint Army-Navy Specifications JAN–P–105, JAN–P–106 and JAN–P–107, respectively. Shipping containers shall be lined with a sealed waterproof bag or its equivalent made from barrier-material conforming to the requirements of Joint Army-Navy Specification JAN–P–125, for case liners. The seams and closures shall have water resistance equal to that of the body material. Shipping containers shall be strapped in accordance with the requirements of the appendix of the applicable box specification. The gross weight shall not exceed approximately 70 pounds.

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G–3. *Marking.*—

G–3a. *Army.*—

G–3a (1). *Marking of interior containers.*—Unless otherwise specified all interior containers shall be marked in accordance with the provisions of Signal Corps Tentative Specification No. 72–6–15.

G–3a (2). *Marking of shipping containers.*—Unless otherwise specified, all shipping containers, both domestic and export, shall be marked in accordance with the provisions of U. S. Army Specification No. 100–2.

G–3b. *Navy.*—

G–3b (1). *Unit packages.*—Unless otherwise specified, unit packages shall be marked with the following information completed:

ARTICLE_____
TYPE DESIGNATION _____

G–3b (2). *Intermediate packages.*—Unless otherwise specified, intermediate packages shall be marked with the following information completed:

ARTICLE_____
QUANTITY_____
TYPE DESIGNATION_____
STOCK NO. _____
CONTRACT NO. _____
MANUFACTURER_____
CONTRACTOR _____

G–3b (3). *Shipping container.*—Unless otherwise specified, shipping containers shall be marked with the following information completed:

ARTICLE_____
QUANTITY_____
TYPE DESIGNATION_____
STOCK NO. _____
CONTRACT NO. _____
CONTRACTOR _____
GROSS WEIGHT _____
CUBE _____

In addition, for domestic shipment only, the consignee's and consignor's addresses shall be given. Marking shall conform to the requirements of the latest issue of the Navy Shipment Marking Handbook.

H. NOTES.

H–1. Requests, requisitions, schedules, and contracts or orders must contain the title, number, and date, of the specification.

H–2. Requests, requisitions, schedules, and contracts or orders should cover the following features:

Style, characteristic, and resistance tolerance of resistors. (See sec. B.)

Type designation. (See par. D–2.)

H–3. *Nonstandard resistors.*—This specification is also intended for

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use in the inspection of nonstandard resistors (par. D–1f), falling within the scope of paragraph B–1.

H–4. This specification is one of a number of such specifications for materials and parts developed jointly between the War and Navy Departments and has been prepared for the purpose of facilitating production of resistors and the equipment in which they are used. This specification should be used in the design of new equipment and, insofar as is practicable, for replacement parts. Additional styles of fixed wire-wound resistors (low power) not covered by this specification are now in use. Existing designs, under which equipment is being and will continue to be fabricated, are such as not to permit complete substitution with resistors conforming to the requirements of this specification, neither are these resistors suitable in all cases for direct replacement of existing resistors.

H–5. Designers of equipment should utilize this specification as extensively as possible in order that maximum production may be had with a minimum waste of time and material.

H–6. *Fungus-proofing.*—The use of materials for external parts that are fungus-proof or fungus-resisting is of increasing importance and should be encouraged. Insufficient data are available on this subject to permit specific recommendations at this time, but supplemental specifications will be issued as knowledge permits.

H–7. This specification is not intended to be restrictive with respect to the details of construction, except where such details are specified. Alternate forms of construction will be considered provided the bid clearly describes the specific points in which the construction proposed differs from that described herein, and provided further that all performance tests and other requirements of this specification are met. When exceptions are not clearly described and indicated as such, it will be assumed that bidders are offering resistors in strict accordance with the requirements of this specification.

H–8. *Qualification (type or brand) approval.*—

H–8a. The right is reserved to reject any bids on brands of resistors which have not been subjected to the required tests and found satisfactory. The attention of the manufacturers is called to this requirement and they are urged to apply for qualification (type or brand approval) tests on samples of resistors which they propose to offer. It is to be understood that the manufacturers shall pay all transportation charges to and from the point where tests are made. In the case of failure of the sample or samples submitted to prove satisfactory, consideration will be given to the requests of the manufacturers for additional tests only after it has been clearly shown that changes have been made in the product with reference to design, methods of manufacture, etc., which the bureau or agency concerned considers sufficient to warrant conducting such additional tests.

H–8a (1). *Army.*—Qualification (type or brand) approval for the Army will be issued by the Signal Corps Standards Agency, 12 Broad Street, Red Bank, N. J. However, the application for submission of test samples should be made to the Army-Navy Electronics Standards Agency (ANESA), 12 Broad Street, Red Bank, N. J.

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H–8a (2). *Navy.*—Qualification (type or brand) approval will be issued by the Bureau concerned, covering resistors which have passed the required tests to the satisfaction of the Navy. However, the application for submission of test samples should be made to the Army-Navy Electronics Standards Agency (ANESA), 12 Broad Street, Red Bank, N. J.

H–8a (3). *Army and Navy.*—Qualification (type or brand) approval covering resistors which have passed the required tests to the satisfaction of the Army and Navy will be issued by the Army-Navy Electronics Standards Agency (ANESA). The application for submission of test samples should be made to the Army-Navy Electronics Standards Agency (ANESA), 12 Broad Street, Red Bank, N. J.

H–8b. *Withdrawal of qualification (type or brand) approval.*—The approving agency reserves the right to make subsequent qualification (type or brand approval) tests on purchased resistors. Any approval granted under this specification will be withdrawn in case of failure under such retest.

H–9. Matters relating to the use of resistors furnished in accordance with this specification for use in the design, production and procurement of Naval radio, radar and sonar equipment on the Electronics Precedence List should be referred to the Bureau of Ships, Electronics Division, Navy Department, Washington 25, D. C.

H–10. *Use notes.*—

H–10a. Where high voltages are present between the resistor circuit and grounded surfaces on which the resistor is mounted, secondary insulation capable of withstanding the voltage condition shall be provided between the resistor and mounting or between the mounting and ground.

H–10b. Care should always be taken in soldering resistors to eliminate any effects caused by soldering irons applied too close to a resistor body or for too long a period.

H–11. Copies of Signal Corps Tentative Specifications may be obtained upon application to the Philadelphia Signal Corps Procurement District, 128 North Broad Street, Philadelphia, Pa. When requesting, refer to specification by both title and identifying symbol.

H–12. Copies of Joint Army-Navy specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies through established War Department channels. Both title and identifying symbol should be stipulated when requesting copies of specifications.

H–13. Copies of Joint Army-Navy specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the

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Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both title and identifying symbol should be stipulated when requesting copies of specifications.

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility or any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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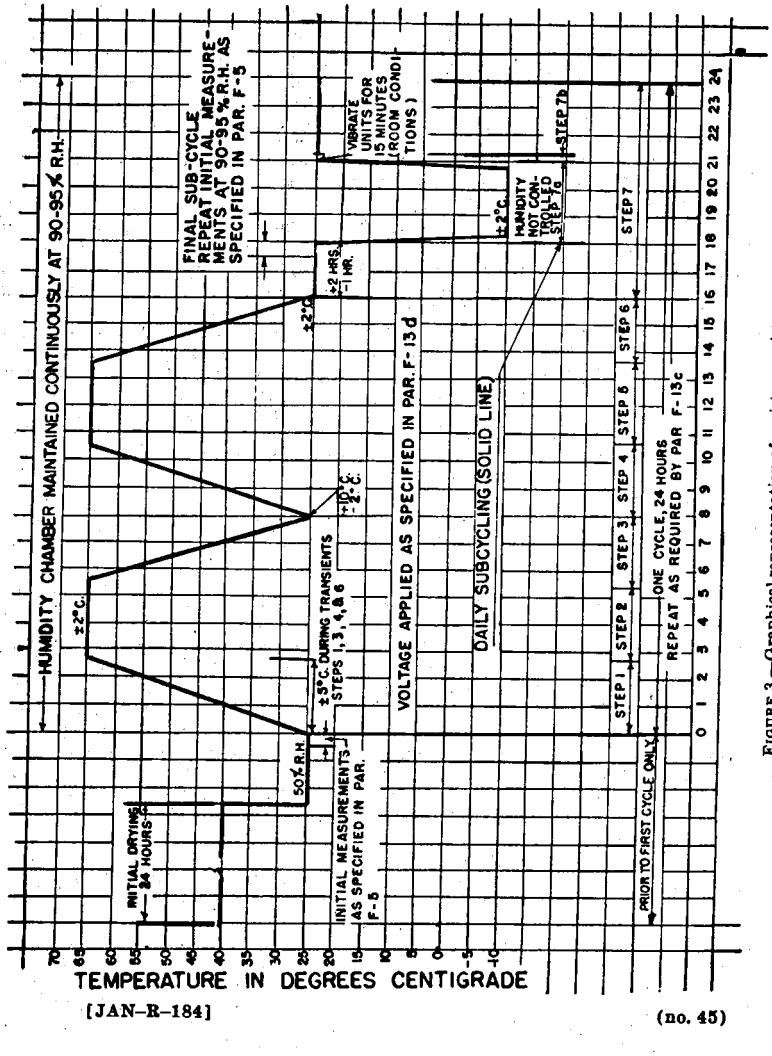
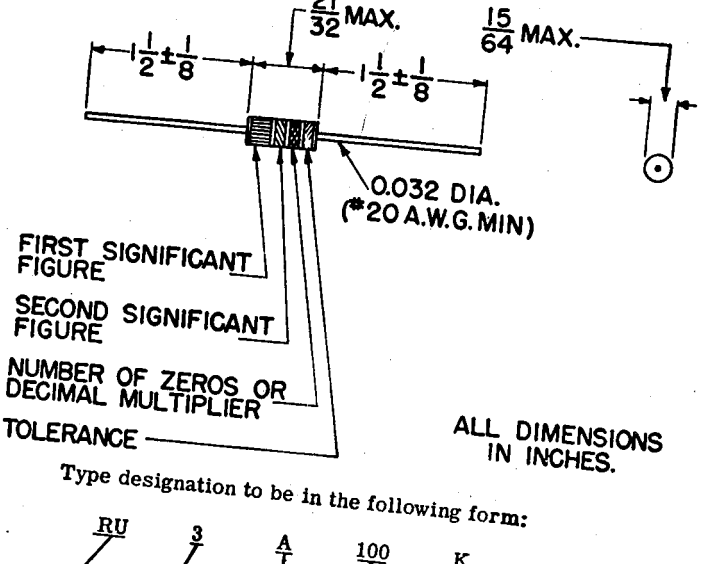
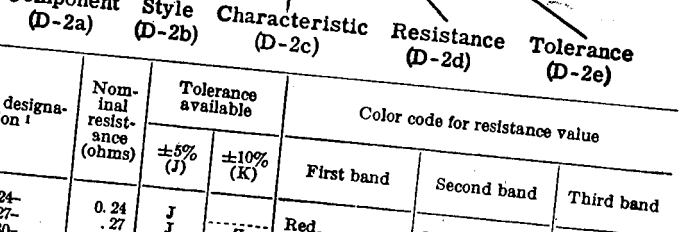


FIGURE 3.—Graphical representation of moisture resistance test.

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Type designation to be in the following form:



Type designation [1]	Nominal resistance (ohms)	Tolerance available		Color code for resistance value		
		±5% (J)	±10% (K)	First band	Second band	Third band
RU3-R24-	0.24	J		Red	Yellow	Silver.
RU3-R27-	.27	J	K	Red	Violet	Do.
RU3-R30-	[2] .30	J		Orange	Black	Do.
RU3-R33-	.33	J	K	Orange	Orange	Do.
RU3-R36-	.36	J		Orange	Blue	Do.
RU3-R39-	.39	J	K	Orange	White	Do.
RU3-R43-	.43	J		Yellow	Orange	Do.
RU3-R47-	.47	J	K	Yellow	Violet	Do.
RU3-R51-	[2] .51	J		Green	Brown	Do.
RU3-R56-	.56	J	K	Green	Blue	Do.
RU3-R62-	.62	J		Blue	Red	Do.
RU3-R68-	.68	J	K	Blue	Gray	Do.
RU3-R75	.75	J		Violet	Green	Do.
RU3-R82-	.82	J	K	Gray	Red	Do.
RU3-R91-	.91	J		White	Brown	Do.
RU3-1R0-	[2] 1.0	J	K	Brown	Black	Gold
RU3-1R1-	1.1	J		Brown	Brown	Do.
RU3-1R2-	1.2	J	K	Red	Red	Do.
RU3-1R3-	1.3	J		Orange	Orange	Do.

[1] Complete type designation will include additional letter symbols to indicate characteristic and resistance tolerance.

[2] For new equipments this value is recommended as a preferred resistance value.

Maximum resistance values are based on use of 0.0015-inch-diameter wire.

FIGURE 4.—Style RU3-½ watt.

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Type designation [1]	Nominal resistance (ohms)	Tolerance available		Color code for resistance value		
		±5% (J)	±10% (K)	First band	Second band	Third band
RU3-1R5-	[2] 1.5	J	K	Brown	Green	Gold
RU3-1R6-	1.6	J		do	Blue	Do.
RU3-1R8-	1.8	J	K	do	Gray	Do.
RU3-2R0-	[2] 2.0	J		Red	Black	Do.
RU3-2R2-	2.2	J	K	do	Red	Do.
RU3-2R4-	2.4	J		do	Yellow	Do.
RU3-2R7-	2.7	J	K	do	Violet	Do.
RU3-3R0-	[2] 3.0	J		Orange	Black	Do.
RU3-3R3-	3.3	J	K	do	Orange	Do.
RU3-3R6-	3.6	J		do	Blue	Do.
RU3-3R9-	3.9	J	K	do	White	Do.
RU3-4R3-	4.3	J		Yellow	Orange	Do.
RU3-4R7	4.7	J	K	do	Violet	Do.
RU3-5R1-	[2] 5.1	J		Green	Brown	Do.
RU3-5R6-	5.6	J	K	do	Blue	Do.
RU3-6R2-	6.2	J		Blue	Red	Do.
RU3-6R8-	[2] 6.8	J	K	do	Gray	Do.
RU3-7R5-	7.5	J		Violet	Green	Do.
RU3-8R2-	8.2	J	K	Gray	Red	Do.
RU3-9R1-	9.1	J		White	Brown	Do.
RU3-100-	[2] 10	J	K	Brown	Black	Black
RU3-110-	11	J		do	Brown	Do.
RU3-120-	12	J	K	do	Red	Do.
RU3-130-	13	J		do	Orange	Do.
RU3-150-	[2] 15	J	K	do	Green	Do.
RU3-160-	16	J		do	Blue	Do.
RU3-180-	18	J	K	do	Gray	Do.
RU3-200-	[2] 20	J		Red	Black	Do.
RU3-220-	22	J	K	do	Red	Do.
RU3-240-	24	J		do	Yellow	Do.
RU3-270-	27	J	K	do	Violet	Do.
RU3-300-	[2] 30	J		Orange	Black	Do.
RU3-330-	33	J	K	do	Orange	Do.
RU3-360-	36	J		do	Blue	Do.
RU3-390-	39	J	K	do	White	Do.
RU3-430-	43	J		Yellow	Orange	Do.
RU3-470-	47	J	K	do	Violet	Do.
RU3-510-	[2] 51	J		Green	Brown	Do.
RU3-560-	56	J	K	do	Blue	Do.
RU3-620-	62	J		Blue	Red	Do.
RU3-680-	[2] 68	J	K	do	Gray	Do.
RU3-750-	75	J		Violet	Green	Do.
RU3-820-	82	J	K	Gray	Red	Do.
RU3-910-	91	J		White	Brown	Do.
RU3-101-	[2] 100	J	K	Brown	Black	Brown.
RU3-111-	110	J		do	Brown	Do.
RU3-121-	120	J	K	do	Red	Do.
RU3-131-	130	J		do	Orange	Do.
RU3-151-	[2] 150	J	K	do	Green	Do.
RU3-161-	160	J		do	Blue	Do.
RU3-181-	180	J	K	do	Gray	Do.
RU3-201-	[2] 200	J		Red	Black	Do.
RU3-221-	220	J	K	do	Red	Do.
RU3-241-	240	J		do	Yellow	Do.
RU3-271-	270	J	K	do	Violet	Do.
RU3-301-	[2] 300	J		Orange	Black	Do.
RU3-331-	330	J	K	do	Orange	Do.
RU3-361-	360	J		do	Blue	Do.
RU3-391-	390	J	K	do	White	Do.
RU3-431-	430	J		Yellow	Orange	Do.
RU3-471-	470	J	K	do	Violet	Do.

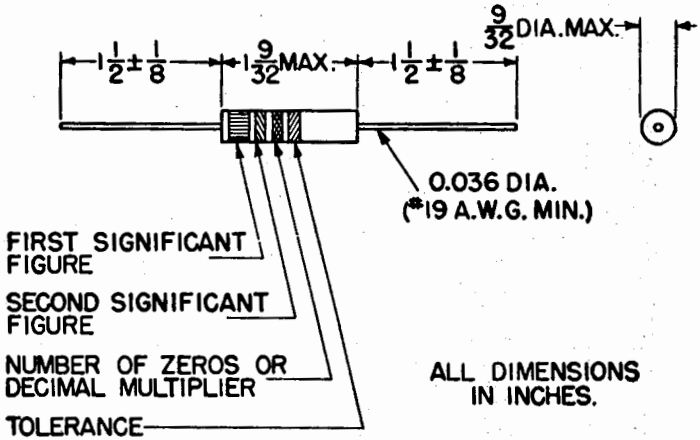
[1] Complete type designation will include additional letter symbols to indicate characteristic and resistance tolerance.

[2] For new equipments this value is recommended as a preferred resistance value.

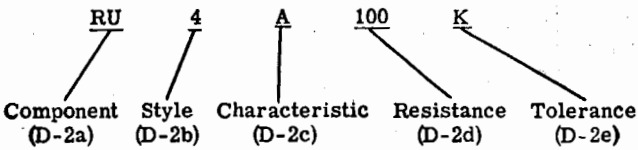
Maximum resistance values are based on use of 0.0015-inch-diameter wire.

FIGURE 4.—Style RU3-½ watt—Continued.

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Type designation to be in the following form:



Type designation [1]	Nominal resistance (ohms)	Tolerance available		Color code for resistance value		
		±5% (J)	±10% (K)	First band	Second band	Third band
RU4-R51-	[2] 0.51	J		Green	Brown	Silver.
RU4-R56-	.56	J	K	do	Blue	Do.
RU4-R62-	.62	J		Blue	Red	Do.
RU4-R68-	.68	J	K	do	Gray	Do.
RU4-R75-	.75	J		Violet	Green	Do.
RU4-R82-	.82	J	K	Gray	Red	Do.
RU4-R91-	.91	J		White	Brown	Do.
RU4-1R0-	[2] 1.0	J	K	Brown	Black	Gold.
RU4-1R1-	1.1	J		do	Brown	Do.
RU4-1R2-	1.2	J	K	do	Red	Do.
RU4-1R3-	1.3	J		do	Orange	Do.
RU4-1R5-	[2] 1.5	J	K	do	Green	Do.
RU4-1R6-	1.6	J		do	Blue	Do.
RU4-1R8-	1.8	J	K	do	Gray	Do.
RU4-2R0-	[2] 2.0	J		Red	Black	Do.
RU4-2R2-	2.2	J	K	do	Red	Do.
RU4-2R4-	2.4	J		do	Yellow	Do.
RU4-2R7-	2.7	J	K	do	Violet	Do.
RU4-3R0-	[2] 3.0	J		Orange	Black	Do.
RU4-3R3-	3.3	J	K	do	Orange	Do.
RU4-3R6-	3.6	J		do	Blue	Do.
RU4-3R9-	3.9	J	K	do	White	Do.
RU4-4R3-	4.3	J		Yellow	Orange	Do.
RU4-4R7-	4.7	J	K	do	Violet	Do.

[1] Complete type designation will include additional letter symbols to indicate characteristic and resistance tolerance.

[2] For new equipments this value is recommended as a preferred resistance value.

Maximum resistance values are based on use of 0.0015 inch diameter wire.

FIGURE 5.—Style RU4, wire wound.

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Type designation [1]	Nominal resistance (ohms)	Tolerance available		Color code for resistance value		
		±5% (J)	±10% (K)	First band	Second band	Third band
RU4-5R1-	[2] 5.1	J		Green	Brown	Gold.
RU4-5R6-	5.6	J	K	do	Blue	Do.
RU4-6R2-	6.2	J		Blue	Red	Do.
RU4-6R8-	[2] 6.8	J	K	do	Gray	Do.
RU4-7R5-	7.5	J		Violet	Green	Do.
RU4-8R2-	8.2	J	K	Gray	Red	Do.
RU4-9R1-	9.1	J		White	Brown	Do.
RU4-100-	[2] 10	J	K	Brown	Black	Black.
RU4-110-	11	J		do	Brown	Do.
RU4-120-	12	J	K	do	Red	Do.
RU4-130-	13	J		do	Orange	Do.
RU4-150-	[2] 15	J	K	do	Green	Do.
RU4-160-	16	J		do	Blue	Do.
RU4-180-	18	J	K	do	Gray	Do.
RU4-200-	[2] 20	J		Red	Black	Do.
RU4-220-	22	J	K	do	Red	Do.
RU4-240-	24	J		do	Yellow	Do.
RU4-270-	27	J	K	do	Violet	Do.
RU4-300-	[2] 30	J		Orange	Black	Do.
RU4-330-	33	J	K	do	Orange	Do.
RU4-360-	36	J		do	Blue	Do.
RU4-390-	39	J	K	do	White	Do.
RU4-430-	43	J		Yellow	Orange	Black.
RU4-470-	47	J	K	do	Violet	Do.
RU4-510-	[2] 51	J		Green	Brown	Do.
RU4-560-	56	J	K	do	Blue	Do.
RU4-620-	62	J		Blue	Red	Do.
RU4-680-	[2] 68	J	K	do	Gray	Do.
RU4-750-	75	J		Violet	Green	Do.
RU4-820-	82	J	K	Gray	Red	Do.
RU4-910-	91	J		White	Brown	Do.
RU4-101-	[2] 100	J	K	Brown	Black	Brown.
RU4-111-	110	J		do	Brown	Do.
RU4-121-	120	J	K	do	Red	Do.
RU4-131-	130	J		do	Orange	Do.
RU4-151-	[2] 150	J	K	do	Green	Do.
RU4-161-	160	J		do	Blue	Do.
RU4-181-	180	J	K	do	Gray	Do.
RU4-201-	[2] 200	J		Red	Black	Do.
RU4-221-	220	J	K	do	Red	Do.
RU4-241-	240	J		do	Yellow	Do.
RU4-271-	270	J	K	do	Violet	Do.
RU4-301-	[2] 300	J		Orange	Black	Do.
RU4-331-	330	J	K	do	Orange	Do.
RU4-361-	360	J		do	Blue	Do.
RU4-391-	390	J	K	do	White	Do.
RU4-431-	430	J		Yellow	Orange	Do.
RU4-471-	470	J	K	do	Violet	Do.
RU4-511-	[2] 510	J		Green	Brown	Do.
RU4-561-	560	J	K	do	Blue	Do.
RU4-621-	620	J		Blue	Red	Do.
RU4-681-	[2] 680	J	K	do	Gray	Do.
RU4-751-	750	J		Violet	Green	Do.
RU4-821-	820	J	K	Gray	Red	Do.
RU4-911-	910	J		White	Brown	Do.
RU4-102-	[2] 1,000	J	K	Brown	Black	Red.
RU4-112-	1,100	J		do	Brown	Do.
RU4-122-	1,200	J	K	do	Red	Do.
RU4-132-	1,300	J		do	Orange	Do.
RU4-152-	[2] 1,500	J	K	do	Green	Do.
RU4-162-	1,600	J		do	Blue	Do.
RU4-182-	1,800	J	K	do	Gray	Do.
RU4-202-	[2] 2,000	J		Red	Black	Do.
RU4-222-	2,200	J	K	do	Red	Do.

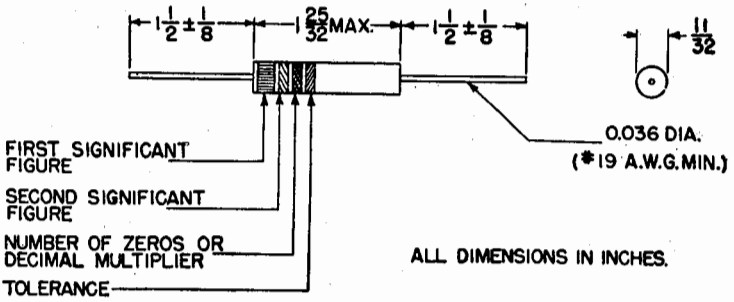
[1] Complete type designation will include additional letter-symbols to indicate characteristic and resistance tolerance.

[2] For new equipment this value is recommended as a preferred resistance value.

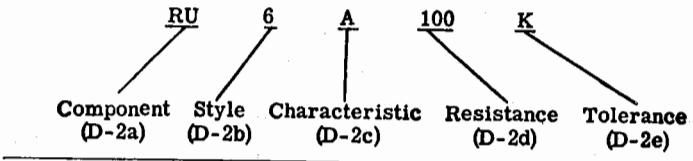
Maximum resistance values are based on use of 0.0015-inch-diameter wire.

FIGURE 5.—Style RU4-1 watt—Continued.

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Type designation to be in the following form:



Type designation [1]	Nominal resistance (ohms)	Tolerance available ±5% (J)	Tolerance available ±10% (K)	Color code for resistance value: First band	Second band	Third band
RU6-1R0-	*1.0	J	K	Brown	Black	Gold.
RU6-1R1-	1.1	J		do	Brown	Do
RU6-1R2-	1.2	J	K	do	Red	Do.
RU6-1R3-	1.3	J		do	Orange	Do.
RU6-1R5-	*1.5	J	K	do	Green	Do.
RU6-1R6-	1.6	J		do	Blue	Do
RU6-1R8-	1.8	J	K	do	Gray	Do.
RU6-2R0-	*2.0	J		Red	Black	Do
RU6-2R2-	2.2	J	K	do	Red	Do.
RU6-2R4-	2.4	J		do	Yellow	Do.
RU6-2R7-	2.7	J	K	do	Violet	Do.
RU6-3R0-	*3.0	J		Orange	Black	Do.
RU6-3R3-	3.3	J	K	do	Orange	Do.
RU6-3R6-	3.6	J		do	Blue	Do.
RU6-3R9-	3.9	J	K	do	White	Do.
RU6-4R3-	4.3	J		Yellow	Orange	Do.
RU6-4R7-	4.7	J	K	do	Violet	Do.
RU6-5R1-	*5.1	J		Green	Brown	Do.
RU6-5R6-	5.6	J	K	do	Blue	Do.
RU6-6R2-	6.2	J		Blue	Red	Do.
RU6-6R8-	*6.8	J	K	do	Gray	Do.
RU6-7R5-	7.5	J		Violet	Green	Do.
RU6-8R2-	8.2	J	K	Gray	Red	Do.
RU6-9R1-	9.1	J		White	Brown	Do.
RU6-100-	*10	J	K	Brown	Black	Black.
RU6-110-	11	J		do	Brown	Do.
RU6-120-	12	J	K	do	Red	Do.

[1] Complete type designation will include additional letter symbols to indicate characteristic and resistance tolerance.

* For new equipments this value is recommended as a preferred resistance value.

Maximum resistance values are based on use of 0.0015-inch diameter wire.

Figure 6.—Style RU6-2 Watt.

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Type designation [1]	Nominal resistance (ohms)	Tolerance available		Color code for resistance value		
		±5% (J)	±10% (K)	First band	Second band	Third band
RU6-130-	13	J		Brown	Orange	Black.
RU6-150-	[2] 15	J	K	do	Green	Do.
RU6-160-	16	J		do	Blue	Do.
RU6-180-	18	J	K	do	Gray	Do.
RU6-200-	[2] 20	J		Red	Black	Do.
RU6-220-	22	J	K	do	Red	Do.
RU6-240-	24	J		do	Yellow	Do.
RU6-270-	27	J	K	do	Violet	Do.
RU6-300-	[2] 30	J		Orange	Black	Do.
RU6-330-	33	J	K	do	Orange	Do.
RU6-360-	36	J		do	Blue	Do.
RU6-390-	39	J	K	do	White	Do.
RU6-430-	43	J		Yellow	Orange	Do.
RU6-470-	47	J	K	do	Violet	Do.
RU6-510-	[2] 51	J		Green	Brown	Do.
RU6-560-	56	J	K	do	Blue	Do.
RU6-620-	62	J		Blue	Red	Do.
RU6-680-	[2] 68	J	K	do	Gray	Do.
RU6-750-	75	J		Violet	Green	Do.
RU6-820-	82	J	K	Gray	Red	Do.
RU6-910-	91	J		White	Brown	Do.
RU6-101-	[2] 100	J	K	Brown	Black	Brown.
RU6-111-	110	J		do	Brown	Do.
RU6-121-	120	J	K	do	Red	Do.
RU6-131-	130	J		do	Orange	Do.
RU6-151-	[2] 150	J	K	do	Green	Do.
RU6-161-	160	J		do	Blue	Do.
RU6-181-	180	J	K	do	Gray	Do.
RU6-201-	[2] 200	J		Red	Black	Do.
RU6-221-	220	J	K	do	Red	Do.
RU6-241-	240	J		do	Yellow	Do.
RU6-271-	270	J	K	do	Violet	Do.
RU6-301-	[2] 300	J		Orange	Black	Do.
RU6-331-	330	J	K	do	Orange	Do.
RU6-361-	360	J		do	Blue	Do.
RU6-391-	390	J	K	do	White	Do.
RU6-431-	430	J		Yellow	Orange	Do.
RU6-471-	470	J	K	do	Violet	Do.
RU6-511-	[2] 510	J		Green	Brown	Do.
RU6-561-	560	J	K	do	Blue	Do.
RU6-621-	620	J		Blue	Red	Do.
RU6-681-	[2] 680	J	K	do	Gray	Do.
RU6-751-	750	J		Violet	Green	Do.
RU6-821-	820	J	K	Gray	Red	Do.
RU6-911-	910	J		White	Brown	Do.
RU6-102-	[2] 1,000	J	K	Brown	Black	Red.
RU6-112-	1,100	J		do	Brown	Do.
RU6-122-	1,200	J	K	do	Red	Do.
RU6-132-	1,300	J		do	Orange	Do.
RU6-152-	[2] 1,500	J	K	do	Green	Do.
RU6-162-	1,600	J		do	Blue	Do.
RU6-182-	1,800	J	K	do	Gray	Do.
RU6-202-	[2] 2,000	J		Red	Black	Do.
RU6-222-	2,200	J	K	do	Red	Do.
RU6-242-	2,400	J		do	Yellow	Do.
RU6-272-	2,700	J	K	do	Violet	Do.
RU6-302-	[2] 3,000	J		Orange	Black	Do.
RU6-332-	3,300	J	K	do	Orange	Do.

[1] Complete type designation will include additional letter symbols to indicate characteristic and resistance tolerance.

[2] For new equipments this value is recommended as a preferred resistance value.

Maximum resistance values are based on use of 0.0015-inch-diameter wire.

FIGURE 6.—Style RU6-2 watt.—Continued

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